HBL Engineering Limited

Rail Signaling Group

In-Process QC Checklist for Stationary Kavach Installation

QC Engineer Name: sushma v1.2

Station ID	Station Name	Zone	Division	Section Name	Initial Date	Updated Date
576765	yg ghh	ECR	Pt Deendayal Upadhy - Pradhankhnta	Pt Deendayal Upadhy - Pradhankhnta	24/03/2026	24/03/2026

Total Points	Closed Points	Open Points
151	109	42

Installation Status	☐ Completed	☒ Not Completed

1.0 Serial Number Verification

S.No	Units	Status	Observation	Image
1.1	Stationary Kavach Unit - 5666666666664	Matching		
1.2	Peripheral Processing Card 1 - 5666666666664	Matching		
1.3	Peripheral Processing Card 2 - 78888888888888	Matching		
1.4	Vital Computer Card -1 - 5444444977	Matching		
1.5	Vital Computer Card -2	Not Applicable		
1.6	Vital Computer Card -3	Not Applicable		
1.7	Voter Card -1	Not Applicable		
1.8	Voter Card -2	Not Applicable		
1.9	Vital Gateway Card 1 (S2S)	Matching		
1.10	Vital Gateway Card 2 (S2S)	Matching		
1.11	Vital Gateway Card 3 (NMS)	Matching		
1.12	EI Gateway-1	Matching		

2.0 Building (New Building Constructed by HBL)	**Not Applicable**

3.0 Tower (New Tower Constructed by HBL)	**Not Applicable**

4.0 Power Supply (Power Supply - DC-DC Converter)	Not Applicable

5.0 Kavach Equipment

S.No	Aspect	Requirement	Status	Observation	Image
5.1 Kavach Unit					
5.1.1	Installation	Kavach unit shall be installed as per the approved floor plan drawing, mounted on the floor using M10 insulators, secured with M10 bolts, and tightened to a torque of 40 Nm as per diagram 5 16 76 0056.	Ok		
5.2 SMOCIP(Station Master's Operation-Cum-Indication Panel)					
5.2.1	Installation	SMOCIP shall be installed in the Station Master's room at an ergonomic height. The panel shall be securely fixed using M6 screws and tightened to a torque of 8 Nm with torque marking applied, as per diagram 5 16 76 0040	Not Applicable		
5.2.2	Termination Unit	SMOCIP Termination Unit shall be wall-mounted near the SMOCIP unit, using insulators, secured with M6 bolts, and tightened to a torque of 8 Nm as per diagram 5 16 76 0046	Not Applicable		
5.2.3	Termination Unit	Power and OFC cables from Kavach termination unit shall be terminated as per diagram 5 16 49 0559.	Not Applicable		
5.2.4	Termination Unit	OFC cables of SMOCIP shall be spliced as per diagram 5 16 49 0559, and proper bunching and routing shall be ensured.	Not Applicable		
5.2.5	Earthing	Unit shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied.	Not Applicable		
5.2.6	Earthing	Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.	Not Applicable		

S.No	Aspect	Requirement	Status	Observation	Image
5.2.7	Functionality	System Health LED shall blink and ensure SYSTEM OK along with the respective station name shall be displayed on the Display.	Not Applicable		
5.2.8	Functionality	Verify that pressing the SOS and Common buttons on the SM-OCIP increments the mechanical counter by one.	Not Applicable		
5.2.9	Checksum	Verify the checksums as per the FAT certificate.	Not Applicable		
5.3 GPS/GSM Antennas					
5.3.1	Installation	Two antennas shall be installed on the Kavach room rooftop with a minimum separation of 3 meters, grouting shall be carried out as per diagram 5 16 67 0039, and torque of 10 Nm shall be applied for M6 fasteners with torque marking provided.	Not Applicable		
5.3.2	Installation	No obstruction above antennas like tree branches, sun-shades, and ensure open to sky etc.	Not Applicable		
5.3.3	Cabling	Antenna cables shall be routed via diverse paths.	Not Applicable		
5.3.4	Cabling	Separate conduits shall be used and Roof conduits shall be sealed against dust, water, and insects.	Not Applicable		
5.3.5	Cabling	In each antenna, the GPS and GSM cables shall be connected to their respective connectors as per the labels provided on the antenna.	Not Applicable		
5.4 RIU(Remote Interface Unit)					
5.4.1	Installation	RIU shall be installed on floor using M10 insulators, secured with M10 bolts, and tightened to a torque of 40 Nm as per diagram 5 16 76 0057.	Not Applicable		
5.4.2	Cabling	All external cables entering into the RIU unit shall pass through cable glands and ensure no cable entry opening shall be used without a cable gland.	Not Applicable		
5.4.3	Cabling	OFC patch cords shall be properly tagged to identify default and standby links.	Not Applicable		

S.No	Aspect	Requirement	Status	Observation	Image
5.4.4	Earthing	Unit shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied.	Not Applicable		
5.4.5	Earthing	Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.	Not Applicable		
5.4.6	FDMS Box installation	FDMS Box shall be installed in the 15U/17U rack of the RIU with proper wall mounting, and OFC splicing shall be carried out as per the network drawing.	Not Applicable		

6.0 RF Communication Equipment On Tower

S.No	Aspect	Requirement	Status	Observation	Image
6.1 RTU(Radio Tower Unit)					
6.1.1	RTU Fixing	Both RTUs shall be firmly secured to the tower platform using M12 bolts and nuts, and a torque of 85 Nm shall be applied as per diagram 5 16 67 0983.	Ok		
6.1.2	RTU Fixing	Ensure RTU doors are fully closed and locked.	Ok		
6.1.3	RTU Earthing	RTU shall be properly earthed by connecting a 35 sq.mm green to GI strip earthing conductor from the RTU earthing bolt to the designated earth pit-4, as per diagram 5 16 76 0043.	Not Applicable		
6.1.4	RTU Earthing	Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.	Ok		
6.1.5	OFC cable termination	OFC cable from the Relay Room shall be spliced and terminated in the splice holder inside the RTU. (Ref. Drawing: 5 16 49 0559)	Ok		
6.1.6	OFC cable termination	OFC cables for RTU shall be spliced as per diagram 5 16 49 0559 and ensure bunching and routing shall be done properly.	Ok		

S.No	Aspect	Requirement	Status	Observation	Image
6.1.7	110V Power cable termination	Cable glands used for 110 V DC power cable entry into RTU shall be firmly tightened.	Ok		
6.1.8	110V Power cable termination	110 V DC power cables shall be terminated inside RTU as per approved drawing 5 16 49 0672.	Ok		
6.1.9	110V Power cable termination	Ensure lugs with sleeves / Ferrules are properly crimped and inserted into the terminal; no loose strands shall be left.	Ok		
6.1.10	Cabling	LMR 600 connection with proper routing; and clamping; No Joints shall be done, as per Tower SOP 5 16 90 0018.	Ok		
6.2 RF Antenna Installation					
6.2.1	RF antenna installation and Audit	RF antenna installation and orientation shall be done as per 10.2dBi omni-directional anteena diagram 5 16 67 0983.	Ok		
6.2.2	RF antenna installation and Audit	RF antenna installation audit report from the installation contractor shall be available in WFMS and there shall be no open points in the audit report.	Ok		

7.0 OFC Networking Rack

S.No	Aspect	Requirement	Status	Observation	Image
7.1	Installation	Network rack shall be installed as per approved floor plan and installation shall be done as per diagram 5 16 76 0058.	Not Applicable		
7.2	Installation	Patch cord routing shall be neat & bend radius to be maintained.	Not Applicable		
7.3	Cabling	All external cables entering the network rack shall pass through the grommets.	Not Applicable		
7.4	Cabling	OFC cables shall be marked using naming tie-tags for easy identification of default and standby links.	Not Applicable		
7.5	Earthing	All networking modules inside the rack shall be connected to the rack chassis using 2.5 sq.mm green/yellow earthing wire.	Not Applicable		

S.No	Aspect	Requirement	Status	Observation	Image
7.6	Earthing	Network rack shall be connected to ring earth using a 10 sq.mm green/yellow earthing wire.	Not Applicable		
7.7	Earthing	Bolts shall be tightened to a torque of 8 Nm, and torque marking shall be applied using yellow paint.	Not Applicable		
7.8	FDMS Box Installation in Network Rack	OFC cables shall be spliced in the FDMS box as per network drawing.	Not Applicable		
7.9	FDMS Box Installation in Network Rack	FDMS boxes shall be clearly marked to identify up and down links.	Not Applicable		
7.10	OFC cable continuity check, after splicing	OTDR test reports shall be available.	Not Applicable		

8.0 Relay Rack (OFC Networking Rack)	**Not Applicable**

9.0 Earthing

S.No	Aspect	Requirement	Status	Observation	Image
9.1	Installation	Earthing shall be done as per RDSO Spec RDSO/SPN/197/2008	Ok		
9.2	Earth Resistance Measurement	Earth resistance shall be measured using a calibrated earth resistance tester. The measured value shall be lessthan or equal to 1 Ohm.	Ok		
9.3	Test Reports	Earth resistance test reports shall be available	Ok		
9.4	Labeling	All earth pits, earthing conductors, and earth points shall be clearly labelled and identifiable.	Available		

10.0 Indoor Wiring / Cabling

S.No	Aspect	Requirement	Status	Observation	Image
10.1	Redundant cabling	Cabling shall be done as per station connectivity diagram 5 16 49 0614 Segregation of power & communication cables.	Ok		
10.2	SMOCIP	12 Core Signaling cable shall be used for button, counter & power supply. OFC armoured cables shall be used for communication.	Ok		
10.3	No Joints	No joints permitted except inside junction boxes or panels.	Ok		
10.4	Color Coding	Colour codes shall be followed for phase, neutral, and earth.	Ok		
10.5	Tagging/Labeling	Cables shall be tagged at both ends.	Ok		
10.6	Termination & C onnection	Proper lugs/ferrules shall be used and terminals tightened correctly.	Ok		

11.0 Outdoor Cabling

S.No	Aspect	Requirement	Status	Observation	Image
11.1	Cable route plan for OFC & Power cable (Relay Room to Tower)	Railway-approved outdoor signalling cable route plan shall be available.	Ok		
11.2	Cable route markers	Route markers shall be installed and clearly visible.	Ok		
11.3	Cable route markers	Where contractually required, underground RFID markers shall be installed and verified.	Ok		

12.0 RFID Tags

S.No	Aspect	Requirement	Status	Observation	Image
12.1	RFID Tag Installation	Ensure the main RFID tag is installed in the direction specified in the RFID tag layout, and a duplicate RFID tag shall be installed at a distance of 3–5 meters from the main tag, except for turnout tags.	Ok		
12.2	RFID Tag Installation	RFID tag mounting shall be firm and secure, ensuring no gap or mechanical play as per RFID tag installation procedure 5 53 76 0055.	Ok		
12.3	RFID Tag data Validation and Placement verification	System-generated RFID data validation report and Placement verification report shall be available.	Available		

13.0 Components Inspection

S.No	Aspect	Requirement	Status	Observation	Image
13.1	Materials Inspection (Which are not Inspected by RDSO or Consignee)	Verify that all subcontractors and HBL personnel use only approved make and part numbers as per the List of Acceptable I&C Materials EG-EN-FT-34.	Ok		

The following are the pending items to be verified during the next inspection

Section Name	Pending Points
Section 1: Serial Number verification	1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.19, 1.20, 1.21, 1.22, 1.23, 1.24, 1.25, 1.26, 1.27, 1.28, 1.29, 1.30, 1.31, 1.32, 1.33, 1.34, 1.35, 1.36, 1.37, 1.38, 1.39, 1.40, 1.41, 1.42, 1.43, 1.44, 1.45, 1.46, 1.47, 1.48, 1.49, 1.50, 1.51, 1.52, 1.53, 1.54, 1.55, 1.56, 1.57, 1.58, 1.59, 1.60, 1.61, 1.62, 1.63, 1.64
Section 4: Power Supply	4.1, 4.2
Section 5: Kavach Equipment	5.1, 5.1.2, 5.1.3, 5.1.4, 5.1.5, 5.1.6, 5.2, 5.3, 5.4
Section 6: RF Communication equipment on tower	6.1, 6.2
Section 7: OFC Networking rack	7.1.1, 7.2.1, 7.3.1, 7.3.2, 7.4.1
Section 8: Relay rack	8.2.1, 8.2.2, 8.4.1
Section 9: Earthing	9.5

Note : This QC report is generated by the system and does not need a physical signature.